

WebP Generator Module



version 1.0.8
by PrestaChamps

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Requirements

Please ensure that your PHP version is at least 5.6 otherwise the module will not work (as mentioned in the product requirements sheet on Addons PrestaShop)

On the server one of the next PHP modules must be configured:

Cwebp => php exec function enabled

Imagick => Imagick php module with webp compiled

Gmagick => Gmagick php module with webp compiled

Gd => Gd php module with imagewebp, imagecreatefrompng and imagecreatefromjpeg functions

Ewww => ewww.io subscription and php curl service

For the module to work properly at least one of the extensions mentioned above (**it is enough for one to be installed and well configured**) needs to be installed and configured on the server.

If the imagick is installed but the regeneration still doesn't work you will need to check if it also supports the webp format. You can check this in php info file. If it does not support the webp format you will need to contact your hosting company and tell them to install the extension with webp support, this will be a **free** service.

Module overview

This module is an essential element if you want a fast website.

WebP Generator module, created for Prestashop sites, will allow users to efficiently regenerate their product, category and manufacturer images thereby increasing the speed of the site.

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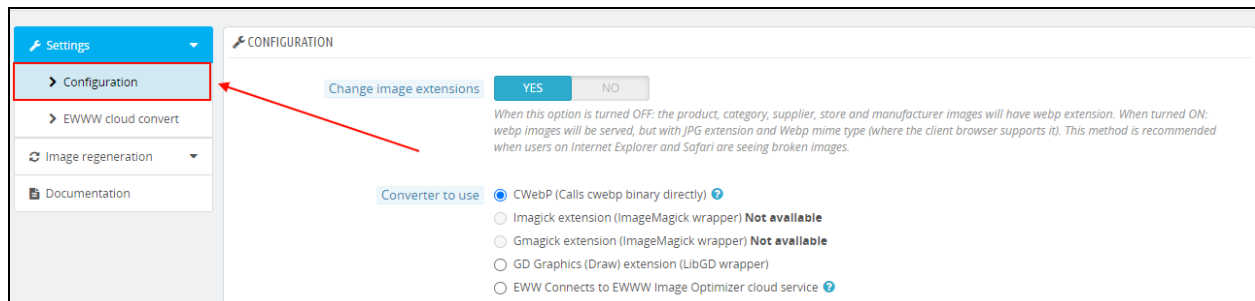
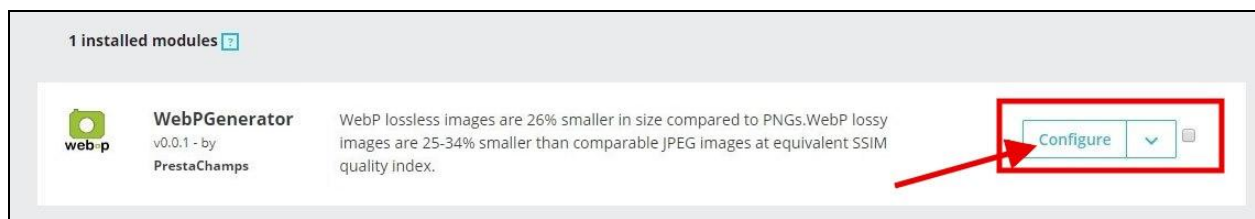
This new technology provides superior lossless and lossy compression for images on your website. After you start generating images through our module it will free up space on your server. Moreover, the quality of your images will not be affected.

The images generated through the **WebP generator module** are 26% smaller in size compared to PNGs. WebP lossy images are 25-34% smaller than comparable JPEG images at equivalent SSIM quality index.

Info: - the module can be installed from the website back office, just like all the other modules.

Module Configuration

- After installation the module can be accessed by clicking on the **Configure** button.



- **Change image extension**

- Choose **“YES”** to serve your image with .jpg or .png extension with Webp mime type

Recommended

- Choose **“NO”** to serve your image with webp extension.

Your users who access the site from Safari or Internet Explorer browser which not supports the webp format, may see broken images.

Change image extensions

YESNO

When this option is turned OFF: the product, category, supplier, store and manufacturer images will have webp extension. When turned ON: webp images will be served, but with JPG extension and Webp mime type (where the client browser supports it). This method is recommended when users on Internet Explorer and Safari are seeing broken images.

Please enable this option during the image generation process. This is recommended to avoid 404 errors of images that have not yet been generated. Once the process is complete you can disable this option.

- **Converter to use** - Set the conversion methods to use depending on your server setup

Converter to use

☒ CWebP (Calls cwebp binary directly) ?

☐ Imagick extension (ImageMagick wrapper) **Not available**

☐ Gmagick extension (ImageMagick wrapper) **Not available**

☐ GD Graphics (Draw) extension (LibGD wrapper)

☐ EWW Connects to EWWW Image Optimizer cloud service ?

If CwebP is available on your server, but the generation of images does not work please try to disable the "Try common system paths" and restart the generation process.

CWEBP Conversion Settings

CWEBP CONVERSION SETTINGS

Try common system paths

YESNO

If CwebP is available on your server, but the generation of images does not work please try to disable the "Try common system paths" and restart the generation process.

Use `nice` command

YESNO

Try supplied binary

YESNO

Turns auto-filter on

YESNO

Command line options

-short

*Read more about all the available parameters here:
https://developers.google.com/speed/webp/docs/cwebp#additional_options*

- **Use `nice` command** - If `nice` command is found on host, binary is executed with low priority in order to save system resources
- **Try common system paths** - This option tests whether cwebp is available in a common system path (eg /usr/bin/cwebp ..)
- **Try supplied binary** - If CWebP is not installed on the server, then supplied binary is selected from Converters/Binaries (according to OS) - after validating checksum
- **Turns auto-filter on** - This algorithm will spend additional time optimizing the filtering strength to reach a well-balanced quality. Unfortunately, it is extremely expensive in terms of computation. It takes about 5-10 times longer to do a conversion. A 1MB picture which perhaps typically takes about 2 seconds to convert, will takes about 15 seconds to convert with auto-filter. So in most cases, you will want to leave this at its default, which is off.
- **Command line options** - This allows you to set any parameter available for cwebp in the same way as you would do when executing cwebp. You could for example set it to `"-sharpness 5 -mt -crop 10 10 40`

Read more about all the available parameters here:

https://developers.google.com/speed/webp/docs/cwebp#additional_options

- **Quality** - Specify the compression factor for RGB channels between 0 and 100. Our recommendation is to set the quality to **90**.
- **Method** - This parameter controls the trade off between encoding speed and the compressed file size and quality. Possible values range from 0 to 6. 0 is fastest. 6 results in best quality.
- **Low memory** - Reduce memory usage of lossy encoding at the cost of ~30% longer encoding time and marginally larger output size.
- **Lossless** - Encode the image without any loss. The option is ignored for PNG's (forced true)
- **Metadata** - Metadata to copy from the input to the output if present.
- **Load WebP polyfill** - Use WebP polyfill to improve old browser compatibility

Quality

70

Method

☐ 1 (best performance)

☐ 2 (good performance)

☐ 3 (decent performance with better quality)

☒ 4 (balance between performance and quality)

☐ 5 (decent quality with better performance)

☐ 6 (high quality)

☐ 7 (best quality)

Low memory

YES

NO

In case of low or limited server resources (specifically memory) enabling this option will reduce memory usage.

Lossless

YES

NO

Recommended to no

Metadata

☐ All

☒ None (recommended)

☐ EXIF

☐ ICC

☐ XMP

Note: Only cwebp supports all values. gd will always remove all metadata. ewww, imagick and gmagick can either strip all, or keep all (they will keep all, unless metadata is set to none)

Load WebP polyfill

YES

NO

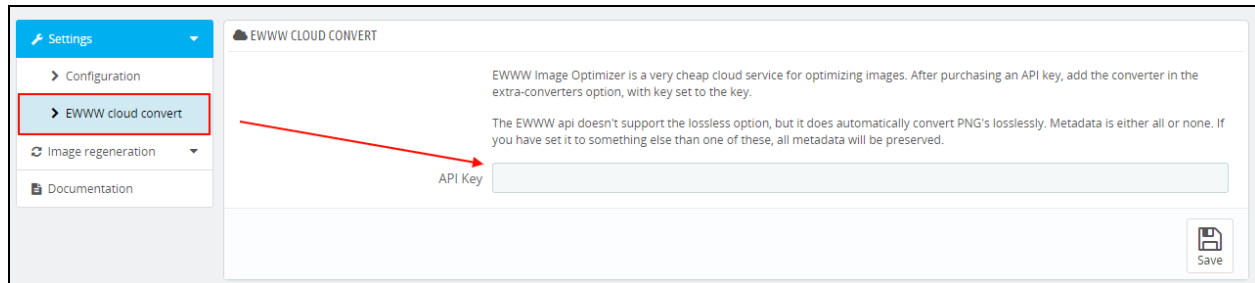
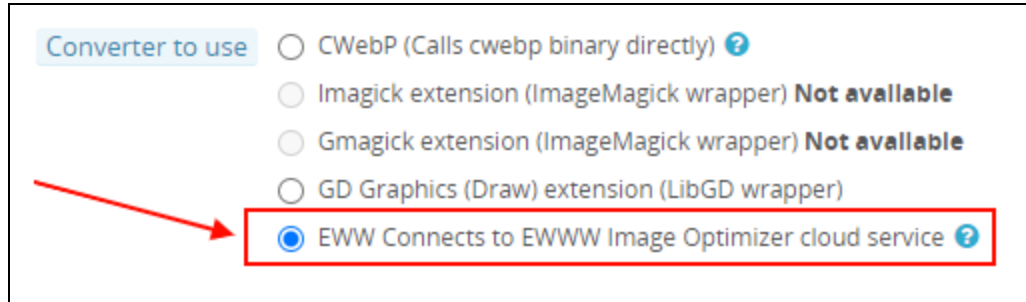
Use WebP polyfill to improve old browser compatibility

EWWW Cloud Convert

EWWW Image Optimizer is a very cheap cloud service for optimizing images. After purchasing an API key, add the converter in the extra-converters option, with the key field containing the API key.

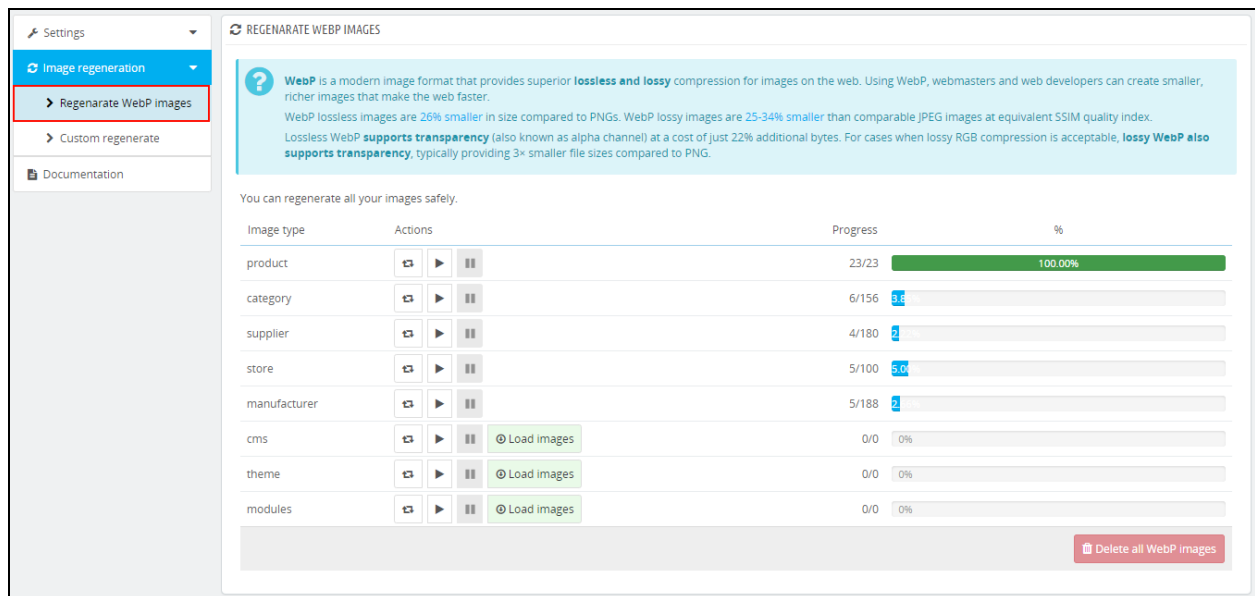
The EWWW API doesn't support the lossless option but it does automatically convert PNGs losslessly. Metadata is either all or none. If you have set it to something else than one of these, all metadata will be preserved.

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Regenerate WebP

After you have configured the module please click on the **Regenerate tab**



You can choose which type of the image you want to generate

- Product Images
- Category Images
- Supplier Images
- Store Images
- Manufacturer Images
- CMS
- Theme
- Modules

Once you've chosen, you can start the generation process. You can see how many images there are for each type and the processing bar will show the percent of the generated images.

REGENERATE WEBP IMAGES

? WebP is a modern image format that provides superior **lossless and lossy** compression for images on the web. Using WebP, webmasters and web developers can create smaller, richer images that make the web faster.

WebP lossless images are **26% smaller** in size compared to PNGs. WebP lossy images are **25-34% smaller** than comparable JPEG images at equivalent SSIM quality index.

Lossless WebP **supports transparency** (also known as alpha channel) at a cost of just 22% additional bytes. For cases when lossy RGB compression is acceptable, **lossy WebP also supports transparency**, typically providing 3× smaller file sizes compared to PNG.

You can regenerate all your images safely.

Image type	Actions	Progress	%
product		23/23	100.00%
category		6/156	3.8
supplier		4/180	2
store		5/100	5.0
manufacturer		5/188	2
cms	Load images	0/0	0%
theme	Load images	0/0	0%
modules	Load images	0/0	0%

Delete all WebP images

You can start generating the type of your images by clicking the **Regenerate all your images**. We implemented the possibility that when you started the image generation process to have the opportunity to pause this process (**just click the Pause the regeneration process**) and to close the tab. When you return, just click on the **“Resume generation process”** and image

generation will continue. This feature is very useful for stores which have got many products and generating all the images can take even a few days.

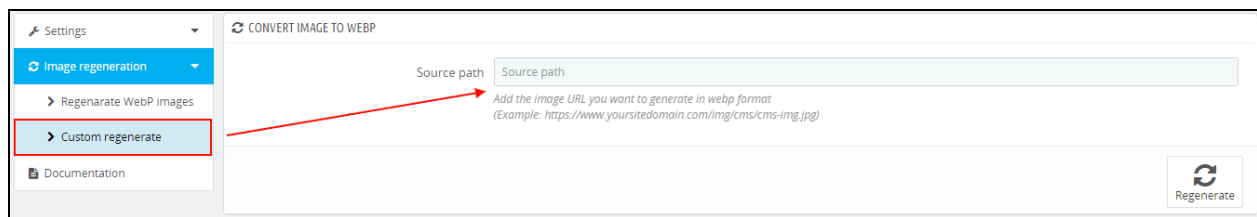
At the CMS, Theme, Modules section before you start the generation process, please press the **“Load images”** button in order to load these type of images.



For the module to load all the images you may have to wait a few seconds, this is the normal behavior of the module.

Custom regeneration

If you want to generate only one image, you need to add the image URL to the **Source path** section and press the **Regenerate** button.



Automatic Generation (Cron Jobs)

At the "Automatic Generation (Cron Jobs)" section you will find the links and the necessary information to set up the cronjob.

This cronjob will run depending on how you set it and periodically will generate all the images that do not have a webp version.

Good to know

WEBP Browser Support:

Currently only Google Chrome and Opera support WebP images. Although other browsers, such as Firefox, Safari or Internet Explorer do not currently support the image format natively, you do not have to worry. Your site's images will not be broken even for clients using these browsers, the images will appear in PNG or JPG format.

WebP lossy support

Google Chrome (desktop) 17+
Google Chrome for Android version 25+
Microsoft Edge 18+
Firefox 65+
Opera 11.10+
Native web browser, Android 4.0+ (ICS)

WebP lossy, lossless & alpha support

Google Chrome (desktop) 23+
Google Chrome for Android version 25+
Microsoft Edge 18+
Firefox 65+
Opera 12.10+
Native web browser, Android 4.2+ (JB-MR1)
Pale Moon 26+

WebP Animation support

Google Chrome (desktop and Android) 32+
Microsoft Edge 18+
Firefox 65+
Opera 19+

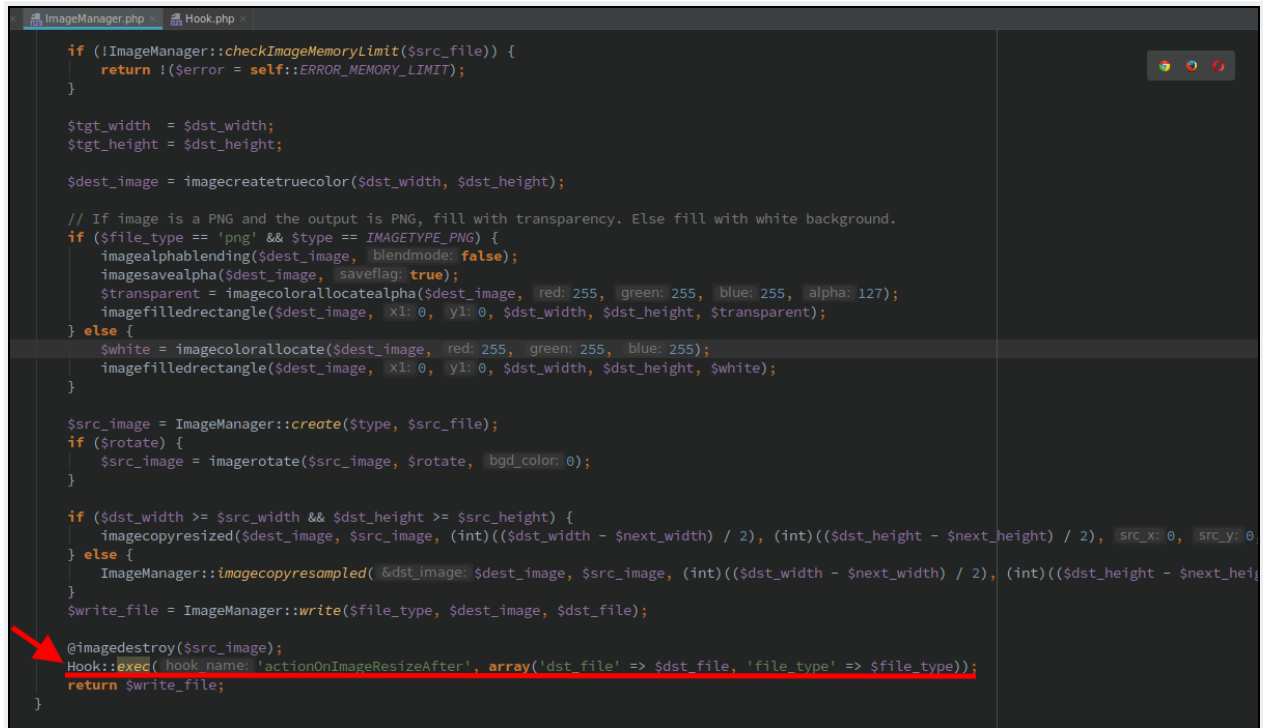
Prestashop 1.6 compatibility

Since Prestashop 1.6 does not execute the `actionOnImageResizeAfter` hook, you need to copy the following code snippet to the `ImageManager` (located in `classes/ImageManager.php`) class `resize` method, before the `return $write_file` statement:

This is how the code should appear after copying the hook:

```
@imagedestroy($src_image);
```

```
Hook::exec('actionOnImageResizeAfter', array('dst_file' => $dst_file, 'file_type' => $file_type));  
return $write_file;
```



```
if (!ImageManager::checkImageMemoryLimit($src_file)) {  
    return !($error = self::ERROR_MEMORY_LIMIT);  
}  
  
$tgt_width = $dst_width;  
$tgt_height = $dst_height;  
  
$dest_image = imagecreatetruecolor($dst_width, $dst_height);  
  
// If image is a PNG and the output is PNG, fill with transparency. Else fill with white background.  
if ($file_type == 'png' && $type == IMAGETYPE_PNG) {  
    imagealphablending($dest_image, blendmode: false);  
    imagesavealpha($dest_image, saveflag: true);  
    $transparent = imagecolorallocatealpha($dest_image, red: 255, green: 255, blue: 255, alpha: 127);  
    imagefilledrectangle($dest_image, x1: 0, y1: 0, $dst_width, $dst_height, $transparent);  
} else {  
    $white = imagecolorallocate($dest_image, red: 255, green: 255, blue: 255);  
    imagefilledrectangle($dest_image, x1: 0, y1: 0, $dst_width, $dst_height, $white);  
}  
  
$src_image = ImageManager::create($type, $src_file);  
if ($rotate) {  
    $src_image = imagerotate($src_image, $rotate, bgd_color: 0);  
}  
  
if ($dst_width >= $src_width && $dst_height >= $src_height) {  
    imagecopyresized($dest_image, $src_image, (int)((($dst_width - $next_width) / 2), (int)((($dst_height - $next_height) / 2), $src_x: 0, $src_y: 0,  
} else {  
    ImageManager::imagecopyresampled($dest_image, $src_image, (int)((($dst_width - $next_width) / 2), (int)((($dst_height - $next_height) / 2), $src_x: 0, $src_y: 0,  
}  
$write_file = ImageManager::write($file_type, $dest_image, $dst_file);  
  
@imagedestroy($src_image);  
Hook::exec('hook_name: 'actionOnImageResizeAfter', array('dst_file' => $dst_file, 'file_type' => $file_type));  
return $write_file;  
}
```

Htaccess Generation:

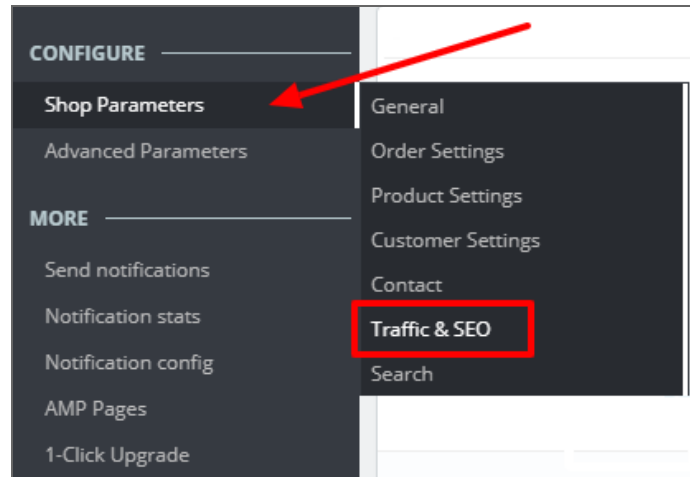
Important!!!

After you have installed the module and generated the images, you have disabled the demo mode. If your images are still not visible on the site means that you have to **generate the htaccess**. This process is very simple and requires a few minutes.

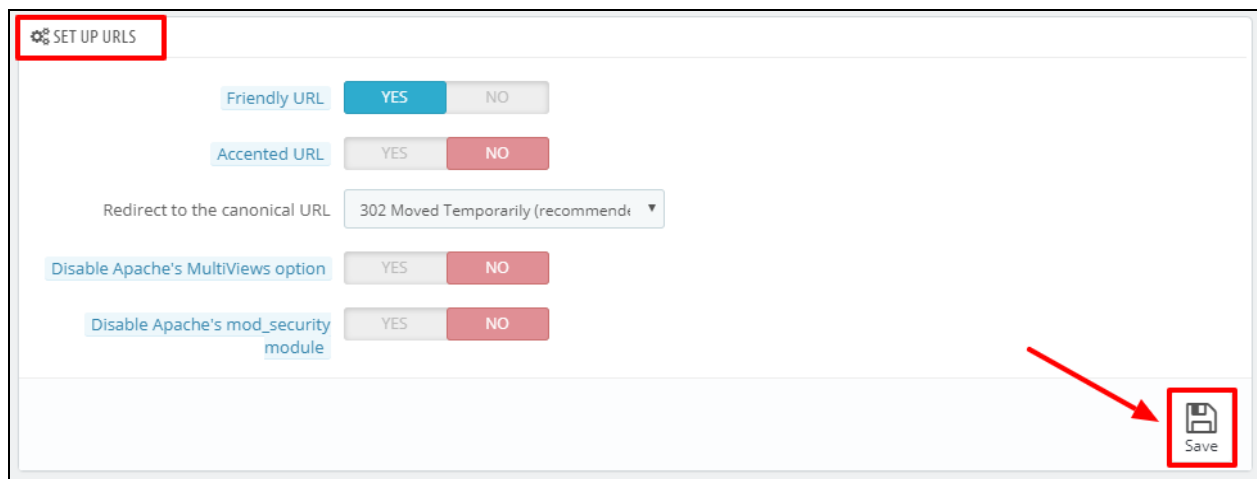
If you use PrestaShop 1.6 please scroll down until the end of the documentation!

PrestaShop 1.7

In the left menu in your back office please go to the **Shop Parameters** section and choose the **Traffic&SEO**

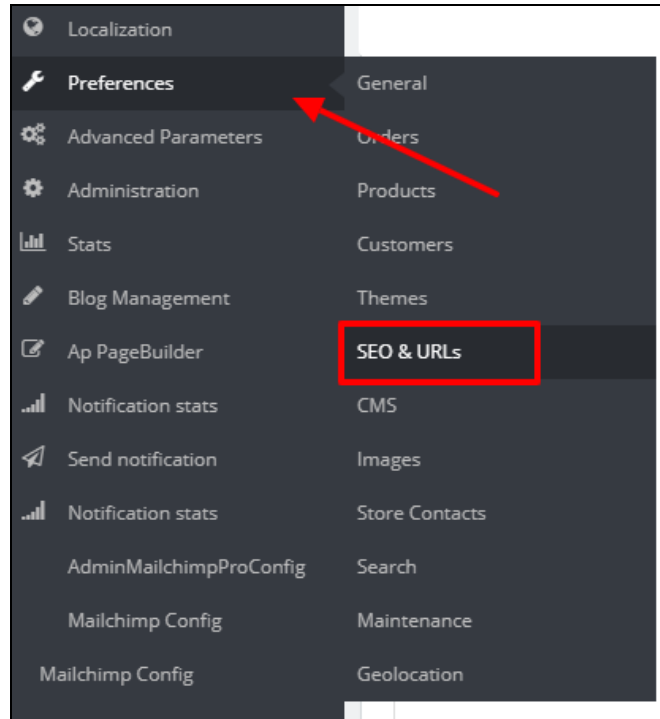


After you have accessed the **Traffic&SEO** section scroll down to the **Set Up URLs** section and without changing anything click on the **Save** button. This process will generate the htaccess.

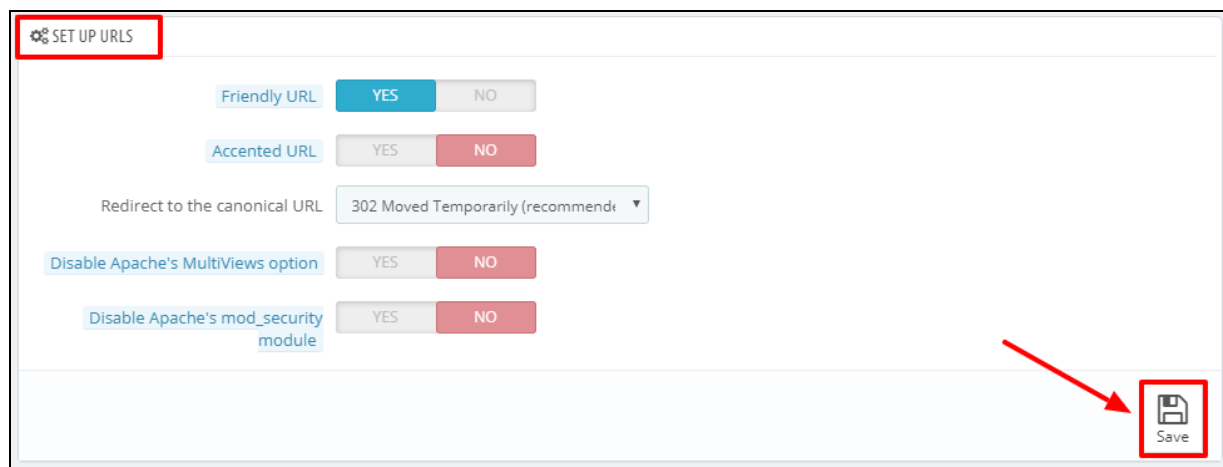


PrestaShop 1.6

In the left menu in your back office please go to the **Preferences** section and choose the **SEO&URLs**



After you have accessed the **SEO&URLs** section scroll down to the **Set Up URLs** section and without changing anything click on the **Save** button. This process will generate the htaccess.



Troubleshoot

1. If your site is under a subfolder the htaccess regeneration will not help to display the images on the site. As the image to work on a site which is in a subfolder you will have to make the following modifications:

Access the FTP, go to the subfolder and edit the htaccess. Look for the next code

WebPGenerator Module - Do not edit
RewriteRule . - [E=REWRITEBASE:/]

after the / sign you have to enter the subfolder name as example:

WebPGenerator Module - Do not edit
RewriteRule . - [E=REWRITEBASE:/subfoldername/]

after you have made this change, you no longer have to generate the htaccess, if you do, you will have to implement this change again.

- 2. The module automatically generates the uploaded images in webp format, but there are cases especially in PrestaShop 1.6 when the images need to be generated manually. When you add, duplicate a product in the back office or you notice that an image of a product is missing, you will have to access the back office, and generate manually the image for that product.**

Go to back office → Product section → Choose the product → Modules Tab → Click on WebP Configure → Regenerate WebP images

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Product Name

Standard product

Basic settings

Combinations

Shipping

Pricing

SEO

Options

Modules

Choose a module to configure

These modules are relative to the product page of your shop.
To manage all your modules go to the [Installed module page](#)


WebPGenerator
1.0.4 by PrestaChamps

WebP lossless images are 26% smaller in size compared to PNGs. WebP lossy images are 25-34% smaller than comparable JPEG images at equivalent SSIM quality index.

CONFIGURE

Product Name

Standard product

en

Sales Product list Help

Basic settings

Combinations

Shipping

Pricing

SEO

Options

Modules


WebPGenerator
1.0.4 by PrestaChamps

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REGENERATE WEBP IMAGES

Module to configure

WebPGenerator

WEBP MODULE CONFIG